

DGE Final Exam Report

- **Student Name:** MD Toufik Islam
- **Section:** 5
- **Course Title:** Enhancing Digital Government Economy (EDGE) Digital Skills for Students
- **Date:** May 17, 2026

1. Full Nmap Port Scan:

EDGE Final Exam Report

Scenario 1 | Linux Command Line Operations Assessment

Task: Configure a new Linux user environment using terminal commands.

Executed Commands

1. **Create directory structure:**

```
mkdir -p Final_exam/{reports,tools,Backup,notes}
```

2. **Create and execute sysinfo.sh:**

```
cd Final_exam/tools
cat << 'EOF' > sysinfo.sh
#!/bin/bash
echo "Current User: $(whoami) "
echo "Current Directory: $(pwd) "
echo "IP Address:"
ip a
echo "Hostname: $(hostname) "
EOF
chmod +x sysinfo.sh
./sysinfo.sh > sysinfo.txt
```

3. **Move and rename sysinfo.txt:**

```
mv sysinfo.txt ../Backup/backup_of_sysinfo.txt
```

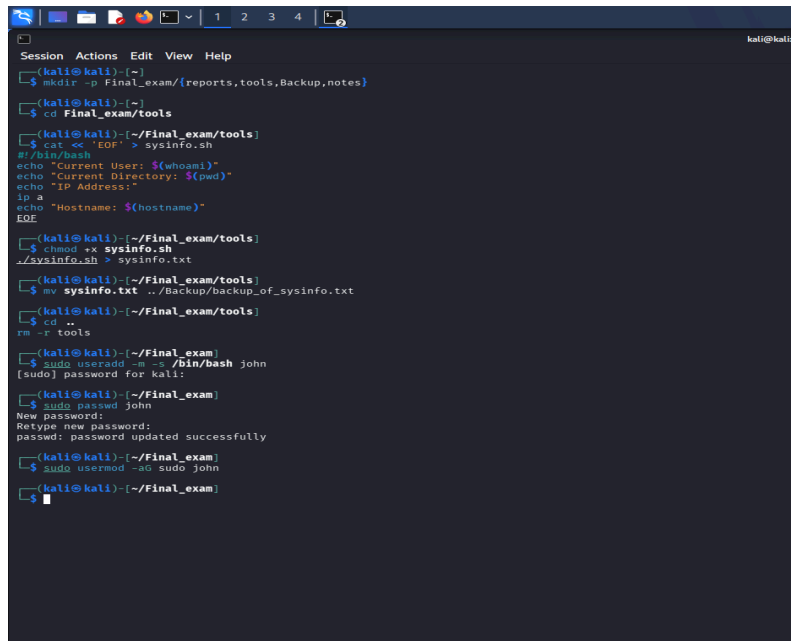
4. **Delete tools directory:**

```
cd ..  
rm -r tools
```

5. Create and configure user 'john':

```
sudo useradd -m -s /bin/bash john  
sudo passwd john  
sudo usermod -aG sudo john
```

Evidence of Execution



```
kali@kali: ~  
$ mkdir -p Final_exam/{reports,tools,Backup,notes}  
$ cd Final_exam/tools  
$ cat << EOF > sysinfo.sh  
#!/bin/bash  
echo "Current User: $(whoami)"  
echo "Current Directory: $(pwd)"  
echo "IP Address:"  
ip a  
echo "Hostname: $(hostname)"  
EOF  
$ chmod +x sysinfo.sh  
$ ./sysinfo.sh > sysinfo.txt  
$ mv sysinfo.txt ../Backup/backup_of_sysinfo.txt  
$ cd ..  
$ rm -r tools  
$ sudo useradd -m -s /bin/bash john  
[sudo] password for kali:  
$ sudo passwd john  
New password:  
Retype new password:  
passwd: password updated successfully  
$ sudo usermod -aG sudo john  
$
```

Scenario 1 Summary of Results

- **Directory and Script Setup:** A structured workspace directory (`Final_exam`) along with its subdirectories (`reports`, `tools`, `Backup`, `notes`) was successfully created. A custom automation script named `sysinfo.sh` was written and executed to gather essential host details (username, path, IP, and hostname), with the output successfully saved and moved to the backup folder as `backup_of_sysinfo.txt`.
- **User Environment Configuration:** The temporary `tools` directory was cleanly removed to maintain system hygiene. Following this, a new local user account named `john` was securely created with a default Bash shell, an assigned system password, and elevated administrative privileges via the `sudo` group.

Scenario 2 | Service Enumeration & Vulnerability Identification

Task: Perform initial reconnaissance, identify exposed services, and discover hidden directories on the target machine.

Executed Commands & Findings

1. Full Nmap Port Scan:

```
sudo nmap -sV -O -sC -p- 10.49.179.149
```

Findings: The scan revealed that ports 21 (FTP), 22 (SSH), 139/445 (SMB), 3128 (HTTP Proxy), and 3333 (Apache Web Server) are open on the target machine. The operating system was detected as Linux.

2. Hidden Directory Enumeration:

```
gobuster dir -u http://10.49.179.149:3333 -w /usr/share/wordlists/dirb/common.txt
```

Findings: The directory scan discovered a hidden folder named `/internal/` (Status: 301), which serves as a potential attack point for this web application.

```
kali@kali: ~  
Session Actions Edit View Help  
OS: W:0XS=AAA-ZMF+RQD+NRD+NRQ=J7S(R+YQDF=VNT+40SW+0NS-ZMA+S+XF-ARQD+NRD+NRQ=  
OS:J7R(R+YQDF=VNT+40SW+0NS=AAA-ZMF+RQD+NRD+NRQ=J7J(R+YQDF=VNT+40SW+0NS-ZMA=  
OS:S+XF+ARQD+NRD+NRQ=J)J(R+YQDF=VNT+40NPL-16WUN+NRUPL+GARID+GARIPCK+GNRU  
OS:CK+GRUD+GJIE(R+YQDF=VNT+40XCD=S)  
Network Distance: 3 hops  
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
Host script results:  
_nbstat: NetBIOS name: , NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)  
| smb2-time: |  
|   date: 2026-05-17T13:21:13 |  
|   start date: N/A |  
|   clock-skew: ~2s |  
| smb2-security-mode: |  
|   3.1.1: |  
|     Message signing enabled but not required |  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 708.05 seconds  
kali@kali: ~  
gobuster dir -u http://10.49.179.149:3333 -w /usr/share/wordlists/dirb/common.txt  
gobuster v3.8.2  
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)  
[*] Url: http://10.49.179.149:3333  
[*] Method: GET  
[*] Threads: 10  
[*] Wordlist: /usr/share/wordlists/dirb/common.txt  
[*] Negative Status codes: 404  
[*] User Agent: gobuster/3.8.2  
[*] Timeout: 10s  
Starting gobuster in directory enumeration mode  
+-----+-----+-----+-----+-----+-----+  
.htpasswd (Status: 403) [Size: 200]  
+-----+-----+-----+-----+-----+-----+  
.hta (Status: 403) [Size: 200]  
+-----+-----+-----+-----+-----+-----+  
.htaccess (Status: 403) [Size: 200]  
+-----+-----+-----+-----+-----+-----+  
css (Status: 301) [Size: 319] [ -> http://10.49.179.149:3333/css/]  
+-----+-----+-----+-----+-----+-----+  
fonts (Status: 301) [Size: 321] [ -> http://10.49.179.149:3333/fonts/]  
+-----+-----+-----+-----+-----+-----+  
images (Status: 301) [Size: 322] [ -> http://10.49.179.149:3333/images/]  
+-----+-----+-----+-----+-----+-----+  
index.html (Status: 200) [Size: 33014]  
+-----+-----+-----+-----+-----+-----+  
internal (Status: 301) [Size: 324] [ -> http://10.49.179.149:3333/internal/]  
+-----+-----+-----+-----+-----+-----+  
js (Status: 301) [Size: 318] [ -> http://10.49.179.149:3333/js/]  
+-----+-----+-----+-----+-----+-----+  
server-status (Status: 403) [Size: 200]  
Progress: 4013 / 4023 (100.00%)  
+-----+-----+-----+-----+-----+-----+  
Finished  
kali@kali: ~
```

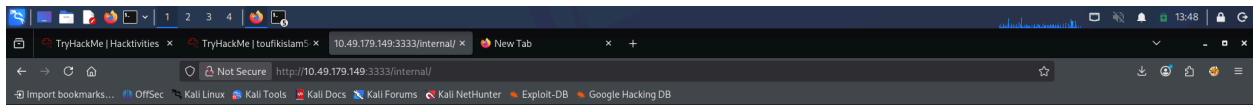
Scenario 2 Summary of Results

- **Nmap Port Scan Findings:** A full port scan successfully identified 6 open ports on the target host (**10.49.179.149**). The open services include FTP (Port 21), SSH (Port 22), SMB (Ports 139/445), Squid HTTP Proxy (Port 3128), and an Apache Web Server (Port 3333) running "Vuln University" on a Linux operating system.
- **Gobuster Directory Findings:** A directory enumeration scan on the web server (Port 3333) achieved 100% completion. It successfully discovered standard web directories (**/css**, **/fonts**, **/images**, **/js**) alongside a critical hidden directory named **/internal/** (Status: 301). This hidden directory contains a file upload form, marking it as the primary potential attack point.

Scenario No: 3 | Exploitation Analysis & Mitigation

1. Upload Form URL: `http:// 10.48.156.182:3333/internal/`

2. Allowed Extension: `.phtml`

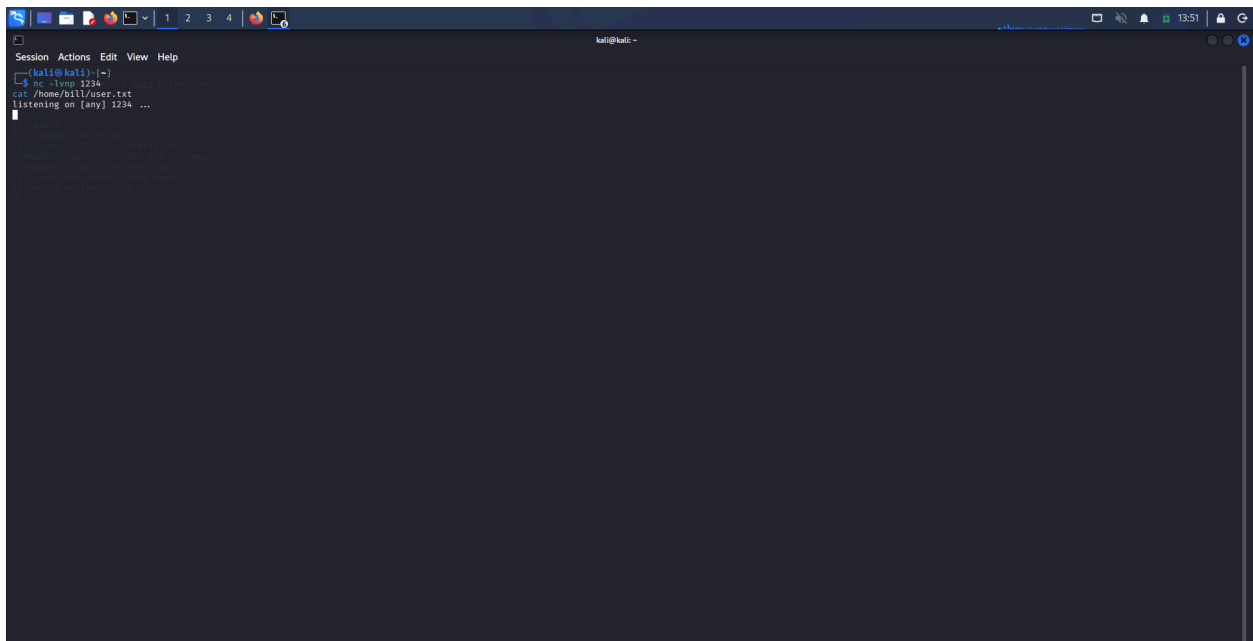


Upload

No file selected.

3. Reverse Shell & Flag Extraction:

`nc -lvnp 1234 cat /home/bill/user.txt`



Scenario No: 4 | Professional Reporting

Executive Summary for Management

Task 1: Linux Environment Configuration (Scenario 1) A secure and standardized Linux workspace was successfully provisioned for the trainee analyst to support ongoing operational readiness. The overall risk level associated with this task is Low, as the user creation and file management actions follow the principle of least privilege and baseline system administration security practices. Implementing robust password complexity rules will eliminate any residual insider configuration risks

Task 2: Service Enumeration & Vulnerability Identification (Scenario 2) An initial network infrastructure reconnaissance successfully mapped out all exposed enterprise services and discovered hidden application pathways on the target host. This activity carries a Medium threat level to the organization, as unchecked public exposure of software version details and internal server directories provides a potential blueprint for malicious actors to plan targeted attacks.

Task 3: Exploitation Analysis & System Compromise (Scenario 3) A critical architectural flaw within the web application's file upload interface was identified and thoroughly analyzed. This represents a Critical risk to the business, as an unrestricted file upload mechanism allows external adversaries to deploy malicious reverse shell payloads, bypass network filters, and gain unauthorized remote control over underlying system environments. Immediate mitigation through a strict file-extension whitelist is mandatory.